

Analysis of Overseas-Owned Property Acquisition Data and Sanctions Matching

Executive Summary

Objective Examine long-term patterns in property acquisitions by overseas owners and assess whether sanctions data could be linked reliably to ownership records.	Data and tools Cleaned Land Registry and sanctions data, SQL joins, regex for consistent formatting, window functions, Jaro-Winkler name similarity scores and Python visualisations.	What this shows Large-scale SQL cleaning, careful entity matching and validation, with clear explanations that do not overstate weak evidence.
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Business Context and Objective

I wanted to establish whether ownership records could be linked to sanctions records well enough to support further investigation into acquisition patterns and possible ownership changes after an entity was placed under sanctions. However, answering this depended on the quality of the data and the information available to connect the records. Property ownership datasets and sanctions lists do not always share a stable identifier. Furthermore, names may be incomplete, inconsistent or written differently across sources, making it harder to establish whether two records refer to the same entity.

I worked mainly in SQL with cleaned Land Registry ownership data, a cleaned sanctions list and a reference table linking sanctions records to companies. I used regex to standardise formatting, joins to connect records, window functions to sequence events and similarity scores to compare names. I then used Python to visualise acquisition patterns over time and across jurisdictions.

Stack: SQL and Snowflake.

Data Sources:

- [UK Sanctions List](#)
- [OCOD \(Overseas Companies Ownership of Data\)](#)
- [Companies House PSC Data](#)

Technical Approach

I began by preparing the data and checking its quality. Before attempting to match records, I checked the structure and completeness of the working ownership table. In the queried sample, the cleaned dataset contained 5,728,514 rows and returned a blank proprietor name rate of 0.00 per cent. These checks gave me confidence to proceed with matching and trend analysis. I then standardised the company registration numbers and proprietor names so that the records could be compared on a consistent basis.

This involved removing unnecessary spaces, using consistent letter case and stripping out unwanted punctuation or other characters that were not letters or numbers. The aim was to reduce avoidable mismatches caused by differences in formatting, so that comparisons reflected the underlying records more closely.

The second stage focused on exact matching. Where a company number or another stable identifier was available, I used it as the preferred way to link records. I standardised company registration numbers in both the sanctions reference and ownership data, then used SQL joins to check whether sanctioned entities could be connected to property titles. An exact identifier match is easier for another analyst to inspect, explain and reproduce than a link based only on similar names. This made it a clear starting point for the project, with an underlying connection that could be checked against the records themselves.

The third stage used fuzzy matching where exact identifiers were missing. This compares how closely names resemble each other, using Jaro-Winkler similarity scoring on the cleaned name fields. Comparing every sanctions name with every proprietor name would have required a large number of comparisons. To keep this manageable, I restricted candidate pairs to names with the same first three letters in their first cleaned token, or name component. I also limited the acceptable difference in name length. These filters are known as blocking. While they narrowed the search enough to make the calculation practical, a sample of 500 sanctions records still generated 758,996 candidate pairs. This shows how quickly the work involved in name matching can grow, even after the search has been restricted.

The fourth stage examined events in sequence. I used the SQL window functions ROW_NUMBER and LEAD to identify later title events occurring after the relevant sanctions designation date. The purpose was to check whether matched entities showed ownership activity that might warrant further regulatory review. The tests returned no confirmed results under the assumptions used. I treated this as a useful analytical finding because it showed what the workflow could establish within those assumptions. It did not provide a confirmed signal of ownership activity after designation.

Table 1. Select Validation Metrics

Rows in cleaned table 5,728,514	Blank proprietor names 0.00%	Fuzzy candidate pairs scored 758,996
Candidates with score >= 85 286	Candidates with score >= 90 0	Highest fuzzy score 88

Results and Interpretation

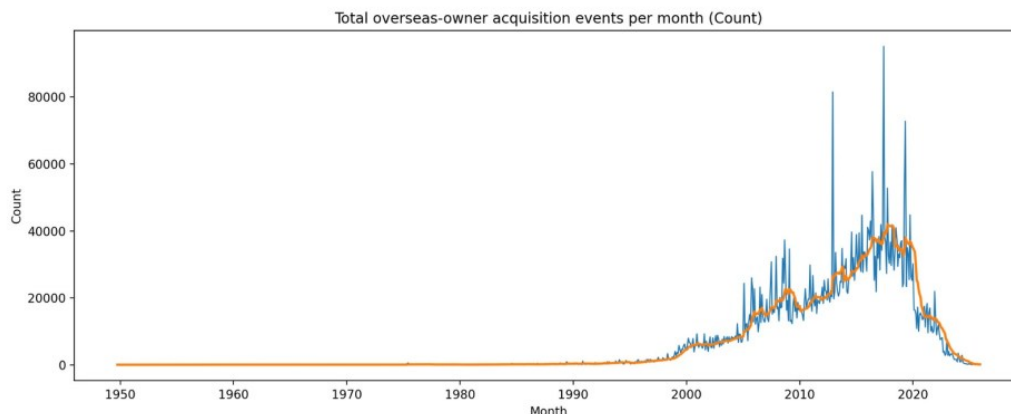
The fuzzy matching results provided a useful finding about the limits of the method.

Only 286 candidate pairs achieved a similarity score of at least 85, none scored 90 or above and the highest score was 88. These results strongly suggested that name similarity alone was insufficient to identify entities confidently in this analysis. I therefore did not treat weak similarities as probable matches. Instead, I used the scores to show how the method had performed, providing a measurable basis for judging the results and explaining why they needed to be treated with caution.

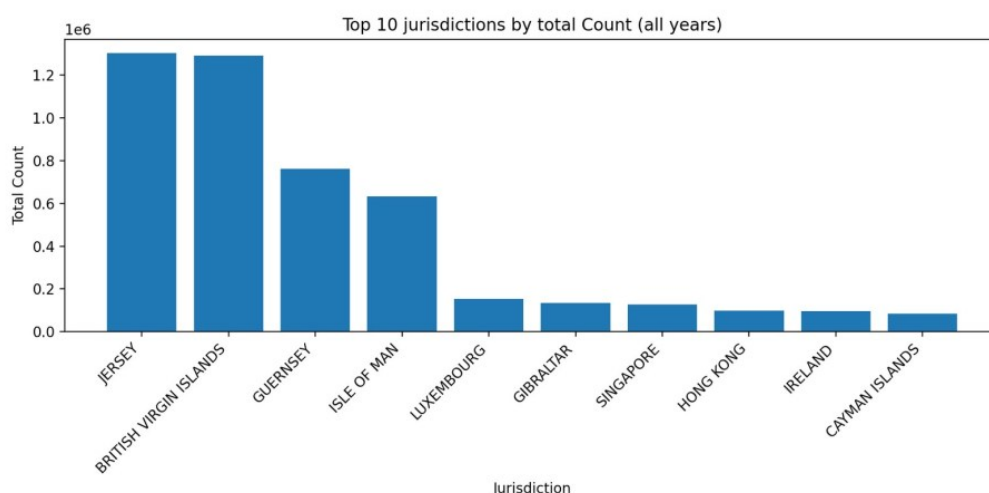
I also used the ownership data to examine acquisition patterns over time. Recorded acquisitions by overseas owners remained low for several decades, increased steadily from the late 1990s and rose more quickly during the 2000s. They reached a peak in the late 2010s before declining sharply after 2020. The jurisdiction totals showed that activity was concentrated in a relatively small number of offshore centres. Jersey and the British Virgin Islands recorded the highest totals, followed by Guernsey and the Isle of Man. Figures 1 and 2 show these patterns over time and across jurisdictions.

This project reinforced the need to be careful about what a possible match can establish. In sanctions analysis, a weak link can create false confidence and may be more harmful than finding no match at all. This is why exact identifiers, clearly stated thresholds and an explanation of uncertainty matter when interpreting the results. Throughout the process, I tracked row counts, checked for missing values and monitored both the number of candidate pairs and the distribution of their scores. These checks helped me assess whether the data and matching process were behaving as expected.

Figures 1 and 2. Visual highlights



Monthly acquisition events built up over time, expanded strongly after 2000 and peaked in the late 2010s before falling sharply after 2020.



Acquisitions were concentrated in a small number of offshore jurisdictions, with Jersey and the British Virgin Islands recording the highest totals.

For further investigation, the most useful outcome was a clearer way to judge the strength of the evidence. A link based on an exact company number gives another analyst records they can inspect and a join they can repeat. A similar name alone does not provide the same basis for confirming identity. This matters in sanctions analysis, where an incorrect match can have serious consequences for compliance and reputation. The project therefore shifted my attention towards the evidence needed to justify further investigation, rather than whether an algorithm could make two names appear similar. This reflects how I approach more complex analysis: a possible connection needs to be supported by evidence that is strong enough for the conclusion being drawn.

Limitations and Next Steps

If I extended the project, I would use aliases, or alternative names, alongside addresses, dates and corporate relationship data to strengthen the matching process. I would also refine the definition of an ownership change and test the workflow on a larger set of sanctioned entities.

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